

SYSTEM HARDENING

1. Objective

The objective of this task was to improve the security posture of the Kali Linux system by:

- Checking for available software updates.
- Applying available system updates.
- Verifying the firewall configuration.
- Auditing currently running services.
- Identifying an unnecessary service.
- Disabling and stopping the unnecessary service.
- Verifying that the service was no longer active or enabled.

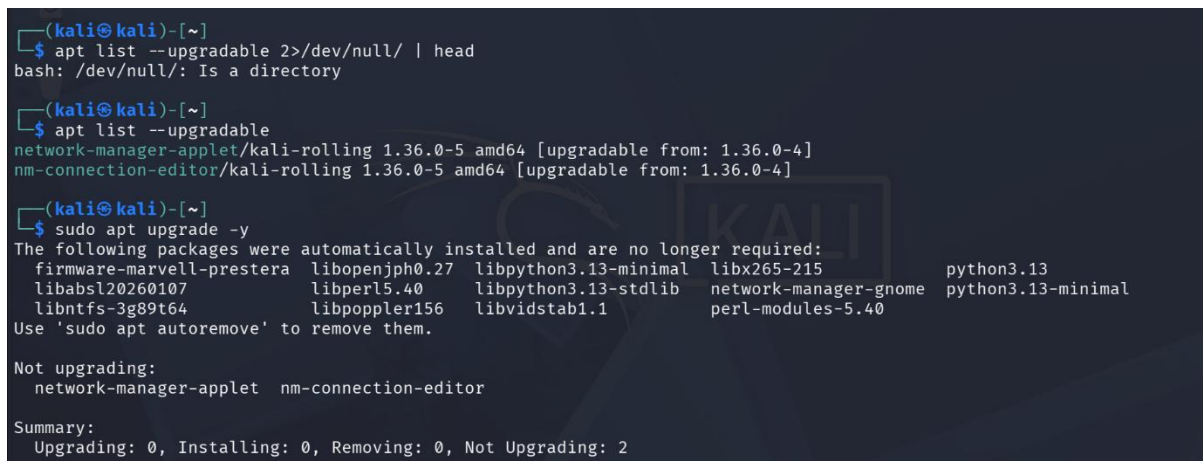
2. Software Update and Patch Management

The system was first checked for available package updates using:

```
apt list --upgradable
```

The command identified available updates, including updates for:

- network-manager-applet
- nm-connection-editor



```
(kali㉿kali)-[~]
$ apt list --upgradable 2>/dev/null/ | head
bash: /dev/null/: Is a directory

(kali㉿kali)-[~]
$ apt list --upgradable
network-manager-applet/kali-rolling 1.36.0-5 amd64 [upgradable from: 1.36.0-4]
nm-connection-editor/kali-rolling 1.36.0-5 amd64 [upgradable from: 1.36.0-4]

(kali㉿kali)-[~]
$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  firmware-marvell-prestera  libopenjph0.27  libpython3.13-minimal  libx265-215  python3.13
  libabsl20260107            libperl5.40  libpython3.13-stdlib  network-manager-gnome  python3.13-minimal
  libntfs-3g89t64           libpoppler156  libvidstab1.1         perl-modules-5.40
Use 'sudo apt autoremove' to remove them.

Not upgrading:
  network-manager-applet  nm-connection-editor

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
```

Figure 1: Available package updates

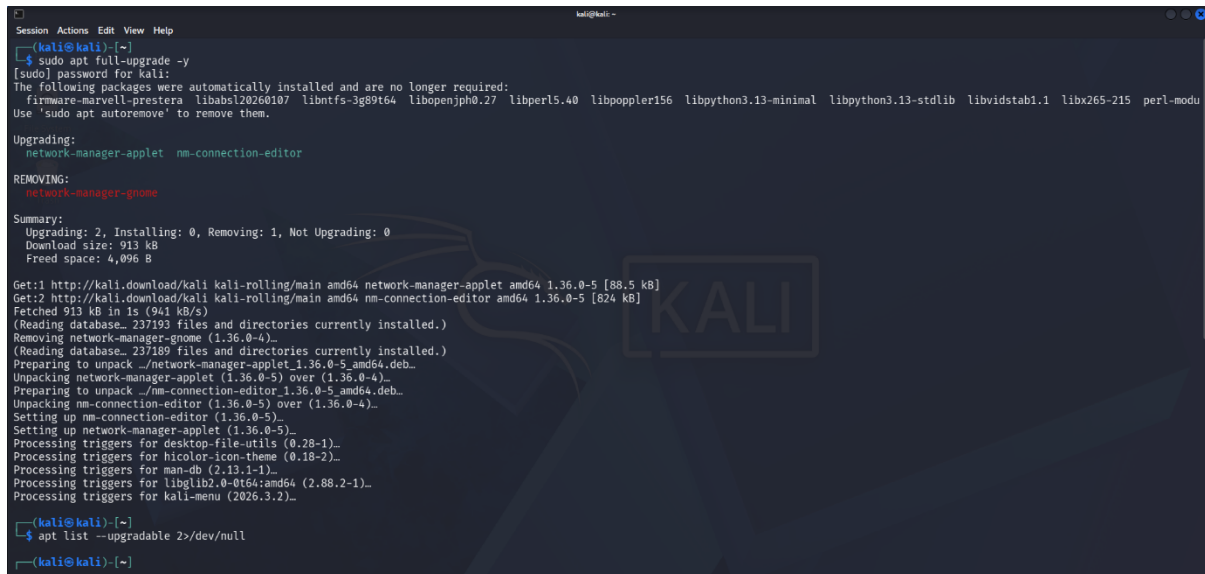
The system was then upgraded using:

```
sudo apt upgrade -y
```

A full upgrade was subsequently performed using:

```
sudo apt full-upgrade -y
```

The upgrade completed successfully. The output showed that two packages were upgraded and one package, `network-manager-gnome`, was removed as part of the package changes.



```
(kali@kali)-[~]
└─$ sudo apt full-upgrade -y
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
  firmware-marvell-presteria libabsl20260107 libbtrfs-3g89t64 libopenjph0.27 libperl5.40 libpoppler156 libpython3.13-minimal libpython3.13-stdlib libvidstab1.1 libx265-215 perl-modu
Use 'sudo apt autoremove' to remove them.

Upgrading:
  network-manager-applet  nm-connection-editor

REMOVING:
  network-manager-gnome

Summary:
  Upgrading: 2, Installing: 0, Removing: 1, Not Upgrading: 0
  Download size: 913 kB
  Freed space: 4,096 B

Get:1 http://kali.download/kali kali-rolling/main amd64 network-manager-applet amd64 1.36.0-5 [88.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 nm-connection-editor amd64 1.36.0-5 [824 kB]
Fetched 913 kB in 1s (941 kB/s)
(Reading database... 237193 files and directories currently installed.)
Removing network-manager-gnome (1.36.0-4).
(Reading database... 237189 files and directories currently installed.)
Preparing to unpack ./network-manager-applet_1.36.0-5_amd64.deb...
Unpacking network-manager-applet (1.36.0-5) over (1.36.0-4)...
Preparing to unpack ./nm-connection-editor_1.36.0-5_amd64.deb...
Unpacking nm-connection-editor (1.36.0-5) over (1.36.0-4)...
Setting up nm-connection-editor (1.36.0-5)...
Setting up network-manager-applet (1.36.0-5)...
Processing triggers for desktop-file-utils (0.28-1)...
Processing triggers for hicolor-icon-theme (0.18-2)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for libgl1:amd64 (2.88.2-1)...
Processing triggers for kali-menu (2026.3.2)...

(kali@kali)-[~]
└─$ apt list --upgradable 2>/dev/null

(kali@kali)-[~]
```

Figure 2: System upgrade completion

A final check was performed using:

```
apt list --upgradable 2>/dev/null
```

No remaining upgrade output was displayed, indicating that there were no packages remaining in the displayed upgrade list.

Security Benefit

Keeping operating-system packages updated helps reduce exposure to publicly known vulnerabilities and ensures that security fixes and software improvements are applied.

3. Firewall Configuration

The status of the UFW firewall was checked using:

```
sudo ufw status verbose
```

The system reported:

```
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
```

```
(kali@kali)-[~]  
$ sudo ufw status verbose  
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), disabled (routed)  
New profiles: skip
```

Figure 3: UFW firewall configuration

The results confirmed that:

- UFW was **active**.
- Firewall logging was enabled at the **low** level.
- Incoming connections were **denied by default**.
- Outgoing connections were **allowed by default**.
- Routed traffic was disabled.

Security Benefit

A default-deny incoming firewall policy reduces the system's exposure to unsolicited network connections. Only services that are explicitly permitted by firewall rules should be reachable from external systems.

4. Running-Service Audit

The running services on the Kali system were enumerated using:

```
systemctl list-units --type=service --state=running
```

The command displayed the services currently running on the system.

The audit identified services including:

- Apache HTTP Server
- MariaDB
- Nessus Vulnerability Scanner
- NetworkManager
- Open VM Tools
- ModemManager
- Systemd services
- Other required system and desktop services

```
(kali@kali)-[~]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

(kali@kali)-[~]
$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
apache2.service                    loaded active running The Apache HTTP Server
colord.service                      loaded active running Manage, Install and Generate Color Profiles
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
haveged.service                    loaded active running Entropy Daemon based on the HAVEGE algorithm
lightdm.service                    loaded active running Light Display Manager
mariadb.service                    loaded active running MariaDB 11.8.8 database server
nessusd.service                    loaded active running The Nessus Vulnerability Scanner
NetworkManager.service             loaded active running Network Manager
open-vm-tools.service               loaded active running Service for virtual machines hosted on VMware
polkit.service                     loaded active running Authorization Manager
rtkit-daemon.service               loaded active running RealtimeKit Scheduling Policy Service
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-networkd.service            loaded active running Network Management
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-udev.service               loaded active running Rule-based Manager for Device Events and Files
systemd-userdbd.service             loaded active running User Database Manager
udisks2.service                    loaded active running Disk Manager
upower.service                     loaded active running Daemon for power management
user@1000.service                   loaded active running User Manager for UID 1000

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

23 loaded units listed.
```

Figure 4: Running-service audit

The purpose of this audit was to identify services that were not required for the system's intended use.

5. Disabling an Unnecessary Service

During the service audit, **ModemManager** was identified as a service that was not required for the laboratory system.

Its status was checked using:

```
systemctl is-active ModemManager.service
```

and:

```
systemctl is-enabled ModemManager.service
```

The final verification produced:

```
inactive
disabled
```

```
(kali㉿kali)-[~]  
$ systemctl is-active ModemManager.service  
inactive  
  
(kali㉿kali)-[~]  
$ systemctl is-enabled ModemManager.service  
disabled
```

Figure 5: ModemManager service disabled and inactive

This confirmed that ModemManager was:

- **Inactive** — it was not currently running.
- **Disabled** — it would not automatically start during boot.

A subsequent service audit showed a reduction in the number of running services from the earlier audit.

Security Benefit

Disabling unnecessary services follows the principle of **least functionality**. Services that are not required increase the system's attack surface and may potentially introduce unnecessary vulnerabilities or network exposure.

Reducing the number of active services therefore helps minimize the opportunities available to an attacker.

6. Hardening Results

Hardening Activity	Result
Checked available updates	Completed
System package upgrade	Completed
Full system upgrade	Completed
Final upgrade check	No remaining displayed upgrades
UFW firewall	Active
Default incoming policy	Deny
Default outgoing policy	Allow
Firewall logging	Enabled
Running-service audit	Completed
ModemManager	Inactive

Hardening Activity	Result
ModemManager boot status	Disabled

7. Security Improvements

The hardening activities resulted in several security improvements:

1. **Patch management**
Available system updates were applied, reducing exposure to known software vulnerabilities.
2. **Firewall protection**
UFW was confirmed to be active with a default-deny incoming policy.
3. **Service minimization**
Running services were reviewed to identify unnecessary functionality.
4. **Reduced attack surface**
ModemManager was disabled because it was not required for the laboratory system.
5. **Verification**
The final service status was checked to confirm that the hardening change had taken effect.

8. Conclusion

The system-hardening exercise successfully improved the security posture of the Kali Linux system.

The system was updated, the UFW firewall was verified as active with incoming traffic denied by default, and running services were audited. An unnecessary ModemManager service was subsequently disabled and verified as inactive.

These measures demonstrate three important system-hardening principles: **regular patching, restrictive network access, and minimizing unnecessary services.**

The completed configuration reduces the system's attack surface and provides a stronger baseline for continued security testing and laboratory activities.